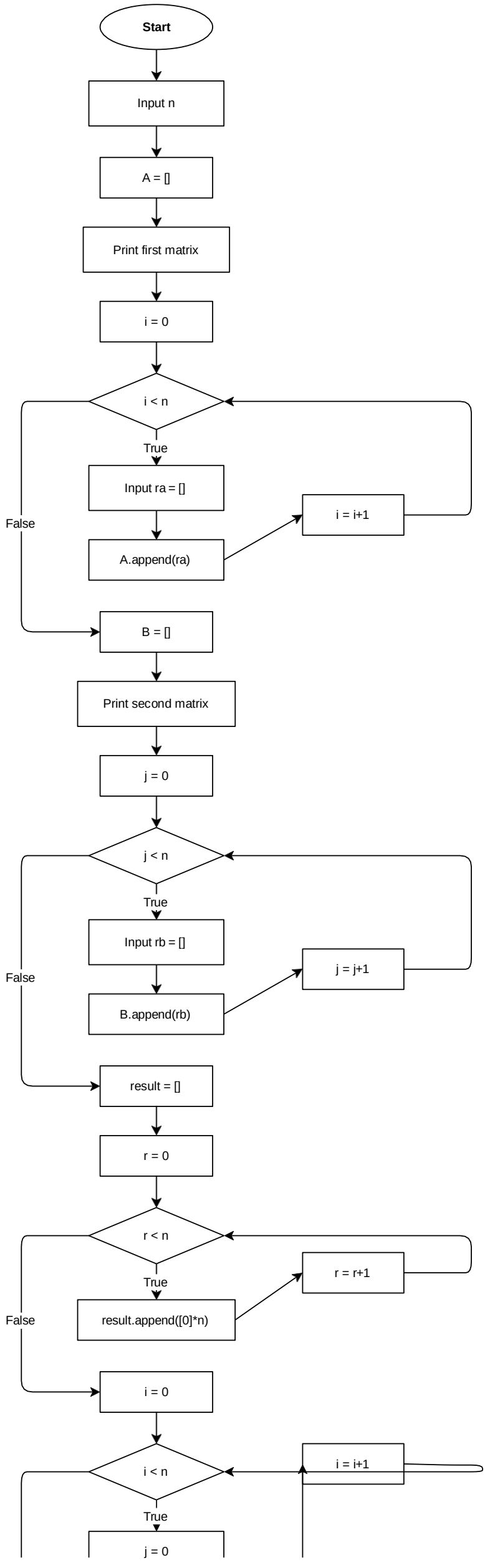
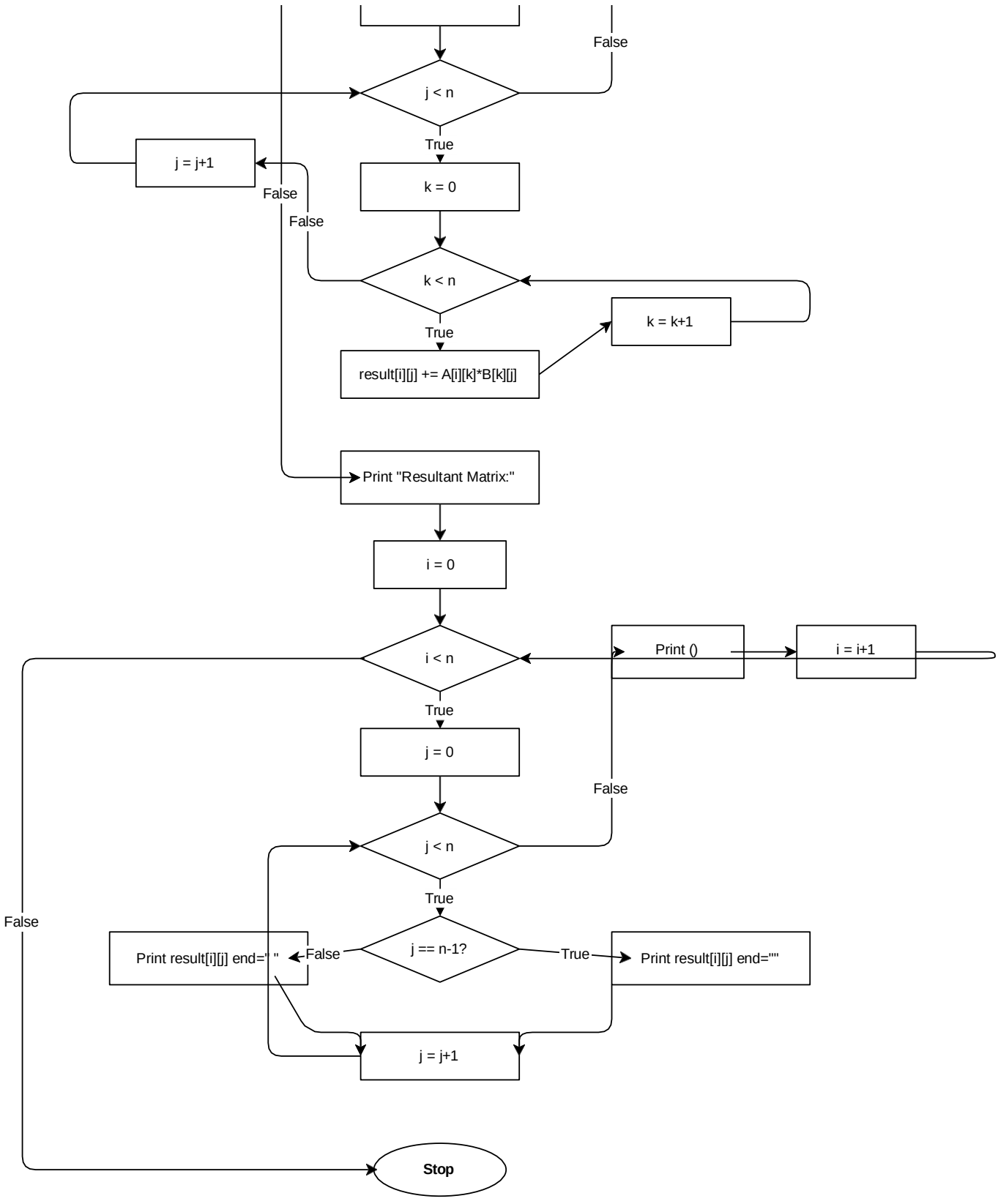
Experiment 8

8.1.1 Multiplication of Two Square Matrix

Algorithm :

```
Step 1: Start
Step 2: Input Dimension (n)
Step 3: A=[]
        Create an empty list A
        Set loop a in range 0 to n:
        Input a row of integers and append to
    A
Step 4: B=[]
        Create an empty list B
        Set loop b in range 0 to n:
        Input a row of integers and append to
    B
Step 5: Result =[]
        Create an empty list result
        Set loop r in range 0 to n:
        Append a list of n zeros to result
Step 6: Multiplication of Matrix
        Set loop i in range 0 to n:
        Set loop j in range 0 to n :
        Set loop k in range 0 to n :
            result[i][j] += (A[i][k] * B[k][j])
Step 7: Print Resultant Matrix
        Set loop i in range $0$ to $n$:
        Set loop j in range $0$ to $n$:
        If j is the last index :
            Print result[i][j] (end="")
        Else:
            Print result[i][j](end=" ")
    Print a new line.
Step 8: Stop
```


Start
Input n
A = []
Print first matrix
i = 0
i < n
True
False
Input ra = []
A.append(ra)
i = i+1
B = []
Print second matrix
j = 0
j < n
True
False
Input rb = []
B.append(rb)
j = j+1
result = []
r = 0
r < n
True
False
result.append([0]*n)
r = r+1
i = 0
i < n
True
i = i+1
j = 0


j < n
False
True
j = j+1
k = 0
False
False
k < n
True
k = k+1
result[i][j] += A[i][k]*B[k][j]
Print "Resultant Matrix:"
i = 0
i < n
Print ()
i = i+1
True
j = 0
False
j < n
True
False
Print result[i][j] end=" "
False
j == n-1?
True
Print result[i][j] end=""
j = j+1
Stop

Code :

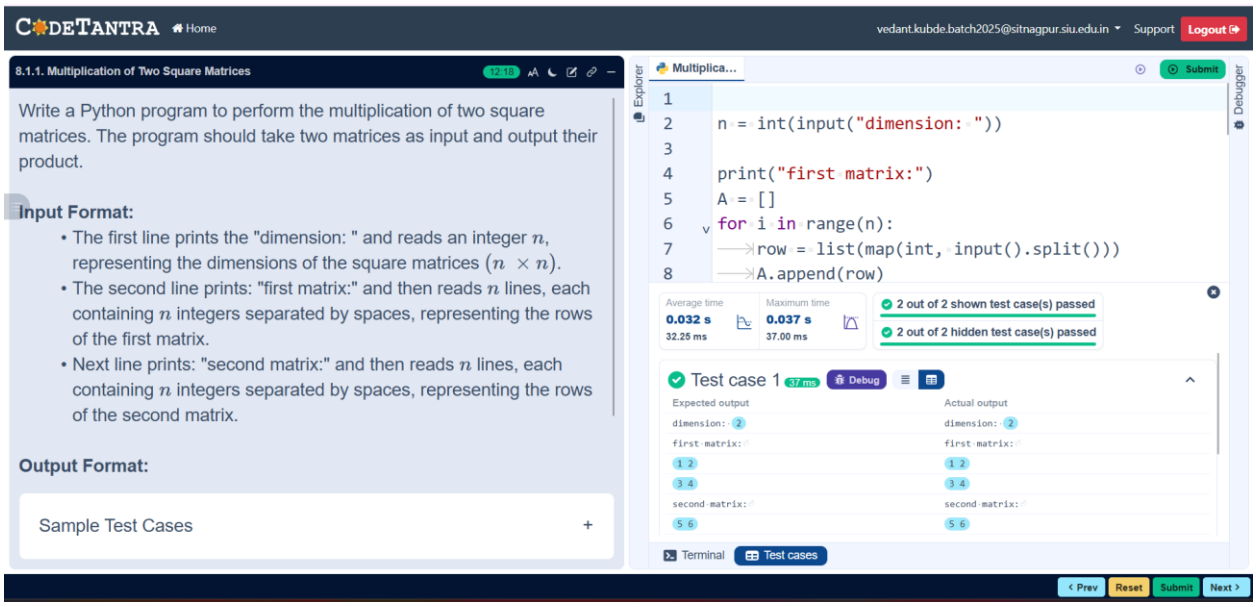
```
n=int(input("dimension: "))
A=[]
print("first matrix:")
for a in range(n):
    ra=list(map(int,input().split()))
    A.append(ra)
B=[]
print("second matrix:")
for b in range(n):
    rb=list(map(int,input().split()))
    B.append(rb)

result=[]
for r in range(n):
    result.append([0]*n)

for i in range(n):
    for j in range(n):
        for k in range(n):
            result[i][j]+=A[i][k]*B[k][j]

print("Resultant Matrix:")
for i in range(n):
    for j in range(n):
        if j==n-1:
            print(result[i][j],end="")
        else:
            print(result[i][j],end=" ")
    print()
```

Execution :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8.1.1. Multiplication of Two Square Matrices
Write a Python program to perform the multiplication of two square matrices. The program should take two matrices as input and output their product.
Input Format:
• The first line prints the "dimension: " and reads an integer n, representing the dimensions of the square matrices $(n \times n)$.
• The second line prints: "first matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the first matrix.
• Next line prints: "second matrix:" and then reads n lines, each containing n integers separated by spaces, representing the rows of the second matrix.
Output Format:
Sample Test Cases
Multiplica...
Submit
n = int(input("dimension: "))
print("first matrix:")
A = []
for i in range(n):
row = list(map(int, input().split()))
A.append(row)
Average time
0.032 s
32.25 ms
Maximum time
0.037 s
37.00 ms
2 out of 2 shown test case(s) passed
2 out of 2 hidden test case(s) passed
Test case 1
Debug
Expected output
Actual output
Terminal
Test cases
Prev
Reset
Submit
Next